HW 8

Monday, 6 March 2023 4:00 PM

Question 1

$A \xrightarrow{k} \text{Products}$, $k(T) = A \exp\left(\frac{-E}{RT}\right)$; c_f, T_f } Surface cond. ; $r = -k[A]$, $k = f(T_f)$

; No external MT resistance

; Heat transfer → only external resistance

(a) Energy balance equation: $\bar{c}_p \frac{\partial T}{\partial t} = \sum_{i}^{m} \rho_b S_{g,i} \alpha_{ij} \hat{r}_i (-\Delta H_i) + \nabla \cdot k_e \nabla T$ → Since we are assuming only external heat transfer resistance → there are no T gradients within the catalyst

Heat flux = Heat generation in the catalyst

(External HT:)

$$h(T - T_f) \cdot A = \int_V \rho_b S_g (-\Delta H) \hat{r}(c,T)\, dV = A \int_0^a \rho_b S_g (-\Delta H) \hat{r}(c,T)\, dx$$

Mass balance equation: $\varepsilon \frac{\partial c_j}{\partial t} = \sum_{k=1}^{5} \nabla \cdot (D_{jk,e} \nabla c_k) + \sum_{i}^{m} \rho_b S_{g,i} \alpha_{ij} \hat{r}_i$

Assumptions: 1) Catalyst surface is "omnipresent" in the pellet → i.e. reaction occurs everywhere in the medium

Rewriting the differential equations in non-dimensional form (for given single reaction):

Conc. PDE: $\varepsilon \frac{\partial c_j}{\partial t} = \sum_{k=1}^{5} \nabla \cdot (D_{jk,e} \nabla c_k) + \sum_{i}^{m} \rho_b S_{g,i} \alpha_{ij} \hat{r}_i$ $\quad \alpha_{ij} = -1$

$\varepsilon \frac{\partial c}{\partial t}^{\to 0} = \nabla \cdot D_e \nabla c - \rho_b S_g \hat{r}(c,T)$ @ SS

$0 = \nabla \cdot D_e \nabla (u \cdot c_f) - \rho_b S_g R(u,v) \cdot \hat{r}(c_f, T_f)$ $\quad D_e$ = constant

$0 = c_f (\nabla \cdot D_e \nabla u) - \rho_b S_g R(u,v) \hat{r}(c_f, T_f)$

$0 = \frac{D_e \cancel{c_f} \nabla^2 u}{\cancel{c_f}} - \rho_b S_g \frac{1}{\cancel{c_f}} k(T_f) \cancel{c_f} R(u,v)$

$0 = \nabla^2 u - \frac{\rho_b S_g k(T_f)}{D_e} \cdot R(u,v)$ ⇒ 1D plate pellet

$\nabla^2 u = \frac{\rho_b S_g k(T_f) R(u,v)}{D_e}$

$\frac{\partial^2 u}{\partial x^2} = \frac{\rho_b S_g k(T_f) R(u,v)}{D_e}$; $x = \rho \cdot a$

$\frac{\partial^2 u}{a^2 \partial \rho^2} = \frac{\rho_b S_g k(T_f)}{D_e} \cdot R(u,v)$

$\frac{\partial^2 u}{\partial \rho^2} = \frac{\rho_b S_g k(T_f) a^2}{} \cdot R(u,v)$; $R(u,v) = \frac{\hat{r}(c,T)}{} = \frac{k(T) \cdot c}{} = u \cdot \frac{A \exp\left(\frac{E}{RT}\right)}{} = u \cdot \exp\left(\frac{-E}{} \left(\frac{}{T_f} - 1\right)\right)$

u $\partial\rho$ νe

$$\frac{\partial^2 u}{\partial \rho^2} = \underbrace{\frac{\rho_b S_g k(T_f) a^2}{D_e}}_{\phi^2} \cdot R(u,\nu) \quad ; \quad R(u,\nu) = \frac{\hat{r}(c,T)}{\hat{r}(c_f,T_f)} = \frac{k(T)\cdot c}{k(T_f)\cdot c_f} = u \cdot \frac{A\exp\left(\frac{E}{RT}\right)}{A\exp\left(\frac{-E}{RT_f}\right)} = u\cdot\exp\left(\frac{-E}{RT_f}\left(\frac{T_f}{T}-1\right)\right)$$

$$R(u,\nu) = u\cdot\exp\left(\gamma\left(1-\frac{1}{\nu}\right)\right)$$

Boundary conditions:

1) @ $\rho = 1$, $u = 1$

2) @ $\rho = 0$, $\frac{\partial u}{\partial x} = 0$ (Bounded conc.)

$$\therefore \quad \frac{\partial^2 u}{\partial \rho^2} = \phi^2 \cdot u \cdot \exp\left(\gamma\left(1-\frac{1}{\nu}\right)\right) = \cdot u$$

$$u = c_1 e^{\sqrt{A}\rho} + c_2 e^{-\sqrt{A}\rho} \quad \Rightarrow \quad 1 = c_1 e^{\sqrt{A}} + c_2 e^{-\sqrt{A}}$$

$$\frac{\partial u}{\partial \rho} = c_1\sqrt{A}\, e^{\sqrt{A}\rho} + c_2(-\sqrt{A})\, e^{-\sqrt{A}\rho}$$

$$0 = c_1\sqrt{A} - c_2\sqrt{A}$$

$$c_1 = c_2$$

$$1 = c_1 e^{\sqrt{A}} + c_1 e^{-\sqrt{A}}$$

$$c_1 = \frac{1}{e^{\sqrt{A}} + e^{-\sqrt{A}}} = \frac{1}{2\cosh(\sqrt{A})}$$

$$\therefore \quad u = c_1\left(e^{\sqrt{A}\rho} + e^{-\sqrt{A}}\right)$$

$$= \frac{1}{2\cosh(\sqrt{A})}\left(2\cdot\cosh(\sqrt{A}\cdot\rho)\right)$$

$$\boxed{u = \frac{\cosh(\sqrt{A}\cdot\rho)}{\cosh(\sqrt{A})}} \quad ; \quad \sqrt{A} = \sqrt{\phi^2\exp\left(\gamma\left(1-\frac{1}{\nu}\right)\right)} = \Psi$$

Solving for temperature:

$$h(T-T_f) = \int_0^a \rho_b S_g(-\Delta H)\,\hat{r}(c,T)\cdot dx \quad ; \quad \hat{r}(c,T) = R(u,\nu)\cdot\hat{r}(c_f,T_f)$$

$$= u\cdot\exp\left(\gamma\left(1-\frac{1}{\nu}\right)\right)\cdot k(T_f)\cdot c_f$$

$$h(\nu\cdot T_f - T_f) = \int_0^1 \rho_b S_g(-\Delta H)\,k(T_f)c_f\cdot u\cdot\exp\left(\gamma\left(1-\frac{1}{\nu}\right)\right)\cdot d(\rho\cdot a)$$

$$(\nu-1) = \int_0^1 u\,\underbrace{\frac{\rho_b\cdot S_g k(T_f)a^2}{D_e}}_{\phi^2}\cdot\underbrace{\frac{c_f\cdot D_e(-\Delta H)}{hT_f\cdot a}}_{\beta}\cdot\exp\left(\gamma\left(1-\frac{1}{\nu}\right)\right)\cdot d\rho \quad \Rightarrow \quad (\nu-1) = \phi^2\beta\int_0^1 R(u,\nu)\cdot d\rho$$

$$(\nu-1) = \phi^2\beta\int_0^1 \exp\left(\gamma\left(1-\frac{1}{\nu}\right)\right)u\cdot d\rho$$

$$(\nu-1) = \phi^2\beta\,\exp\left(\gamma\left(1-\frac{1}{\nu}\right)\right)\int_0^1 u\cdot d\rho$$

$$(\nu-1) = \phi^2\beta\,\exp\left(\gamma\left(1-\frac{1}{\nu}\right)\right)\int_0^1 u\cdot d\rho \quad ; \quad \Psi = \phi\exp\left(\frac{\gamma}{2}\left(1-\frac{1}{\nu}\right)\right)$$

$$\Psi^2 = \phi^2\exp\left(\gamma\left(1-\frac{1}{\nu}\right)\right)$$

$$(\nu-1) = \beta\Psi^2\int_0^1 u\cdot d\rho$$

$$(\nu - 1) = \beta \psi^2 \int_0^1 \frac{\cosh(\psi p)}{\cosh(\psi)} \cdot dp$$

$$(\nu - 1) = \frac{\beta \psi^2}{\cosh(\psi)} \frac{\sinh(\psi)}{\psi}$$

$$\boxed{(\nu - 1) = \beta \psi \tanh(\psi)} \quad ; \quad \psi = \phi \exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right)$$

$$\eta = \frac{r_{obs}}{\hat{r}_s} = \frac{\int_V r \, dV}{r_s V} = \frac{\int_V \frac{r}{r_s} \cdot A \, da}{A \cdot a} = \int_0^1 R(u,v) \cdot dp \quad ;$$

$$\boxed{\eta = \frac{(\nu - 1)}{\phi^2 \beta}}$$

(b) Limiting case:

+ Small ϕ: $\phi \to 0$; $(\nu - 1) = \beta \left(\phi \exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right)\right) \tanh\left(\phi \exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right)\right)$ ($\phi \to 0$)

$\nu - 1 = 0 \Rightarrow \boxed{\nu \to 1}$

$$\eta = \frac{\psi \tanh(\psi)}{\phi^2} = \frac{\phi \exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right) \tanh\left(\phi \exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right)\right)}{\phi^2} \approx \frac{\tanh(\phi)}{\phi} \approx 1 \; ; \quad \boxed{\eta \to 1}$$

+ Large ϕ: $\phi \to \infty$; $\boxed{\eta \to 0}$

$(\nu - 1) = \beta \left(\phi \exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right)\right) \tanh\left(\phi \exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right)\right)$ ($\tanh \to 1$)

$$\frac{(\nu - 1)}{\exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right)} = \beta \phi$$

$\boxed{\nu \to 1}$

(C) $\eta = f(\nu, \beta, \phi)$; $\nu = f(\psi, \phi)$

$(\nu - 1) = \beta \psi \tanh \psi$

$(\nu - 1) = \beta \phi \exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right) \tanh\left(\phi \exp\left(\frac{\gamma}{2}\left(1 - \frac{1}{\nu}\right)\right)\right)$

↓

Solving for ν for given ϕ values → plug in η & plot η vs. ϕ